

Extracellular granzyme A, complexed to proteoglycans, is protected against inactivation by protease inhibitors.

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Granzyme A (GrA) and B (GrB) together with perforin are the main constituents of cytotoxic granules of cytotoxic T lymphocytes (CTLs) and natural killer (NK) cells. The cytotoxic proteins are released to deliver a lethal hit during contact between the CTL or NK cell and target cell. With the use of an enzyme-linked immunosorbent assay for antigenic levels, we showed in a recent study that plasma of patients with activated CTLs and NK cells contain elevated levels of extracellular GrA. In this study, we determined the form and proteolytic capacity of this extracellular GrA detected in plasma. With the use of various assays, we show that part of the extracellular GrA circulates in the mature conformation and is bound to proteoglycans that protect it against inactivation by protease inhibitors, such as antithrombin III and alpha-2-macroglobulin, whereas another part of GrA circulates as a complex with antithrombin III. Finally, with the use of a novel assay for active GrA, we demonstrate that some plasma samples with high levels of extracellular GrA contain active GrA. These results suggest that various forms of extracellular GrA occur in vivo and that the regulation of GrA activity may be modified by proteoglycans. These data support the notion that granzymes may exert extracellular functions distant from the site of CTL or NK cell interaction with their target cells. (Blood. 2000;95:1465-1472)

Synaptic secretion from human natural killer cells is diverse and includes supramolecular attack particles.

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Natural killer (NK) cells form immune synapses to ascertain the state of health of cells they encounter. If a target cell triggers NK cell cytotoxicity, lytic granules containing proteins including perforin and granzyme B, are secreted into the synaptic cleft inducing target cell death. Secretion of these proteins also occurs from activated cytotoxic T lymphocytes (CTLs) where they have recently been reported to complex with thrombospondin-1 (TSP-1) in specialized structures termed supramolecular attack particles (SMAPs). Here, using an imaging method to define the position of each NK cell after removal, secretions from individual cells were assessed. NK cell synaptic secretion, triggered by ligation of NKp30 or NKG2D, included vesicles and SMAPs which contained TSP-1, perforin, and granzyme B. Individual NK cells secreted SMAPs, CD63+ vesicles, or both. A similar number of SMAPs were secreted per cell for both NK cells and CTLs, but NK cell SMAPs were larger. These data establish an unexpected diversity in NK cell synaptic secretions.

Granule serine proteases are normal nuclear constituents of natural killer cells.

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One mechanism by which cytotoxic T lymphocytes and natural killer cells inflict target cell death depends upon secreting the contents of their specialized cytoplasmic granules, containing a pore-forming protein, perforin, and a family of homologous serine proteases ("granzymes") with various enzyme activities. We used a granzyme B-specific mouse anti-human monoclonal antibody 2C5 and Western blotting to demonstrate that nuclear extracts of human interleukin-2-stimulated peripheral blood mononuclear cells,

the human NK leukemia cell line YT, and the rat NK leukemia cell line RNK-16 contain abundant granzyme B. In interleukin-2-activated peripheral blood mononuclear cells, more than 50% of the total cellular granzyme B was present in the nuclear lysate. Nuclear granzyme B had an apparent molecular mass of approximately 32 kDa in human cells and approximately 30 kDa in RNK-16 and was eluted from immobilized heparin at the same NaCl concentration as granzyme B from cytoplasmic granules. Granzyme B that was affinity-purified with 2C5 from the nuclei of YT or human LAK cells was capable of efficiently cleaving synthetic peptide thiobenzyl ester substrates with the same specificity (peptide cleavage after aspartic acid) as granule-localized granzyme B. By contrast perforin, which colocalizes with granzymes in cytotoxic granules, was not detectable in nuclear lysates. Granzyme B was also demonstrated to be present in the nucleus and cytoplasmic granules of YT by immunohistochemical staining with monospecific anti-granzyme B antisera. Other protease activities (tryptase and peptide cleavage after methionine) were also readily detectable in nuclear and cytoplasmic lysates of YT, RNK-16, and LAK cells, as determined by the cleavage of the synthetic substrates N alpha-benzyloxycarbonyl-L-lysine thiobenzyl ester (BLT) and Boc-Ala-Ala-Met-S-benzyl, except that BLT-esterase activity was absent from the nucleus of YT. The localization of serine proteases in the nucleus was restricted to lymphocytes with cytotoxic capacity, as non-cytotoxic cell lines expressed high levels of peptide cleavage after methionine and tryptase activities in their cytoplasm, but possessed no nuclear serine protease activity. Furthermore, non-cytotoxic monkey kidney COS-7 cells transfected with an SV40-driven expression plasmid incorporating full-length human granzyme B cDNA contained abundant cytoplasmic granzyme B, but demonstrated minimal nuclear granzyme B accumulation. We conclude that serine proteases of NK cells are not restricted to cytolytic granules and, further, that their capacity to access the nucleus may have implications for the role of these enzymes in eliciting target cell death.

Perforin and granzymes in neurological infections: From humans to cattle.

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Perforin and granzymes are essential components of the cytotoxic granules present in cytotoxic T lymphocytes and natural killer cells. These proteins play a crucial role in a variety of conditions, including viral infections, tumor immune surveillance, and tissue rejection. Besides their beneficial effect in most of these situations, perforin and granzymes have also been associated with tissue damage and immune diseases. Moreover, it has been reported that perforin and granzymes released during viral infections could contribute to the pathogenesis of diseases. In this review, we summarize the information available on human perforin and granzymes and their relationship with neurological infections and immune disorders. Furthermore, we compare this information with that available for bovine and present data on perforin and granzymes expression in cattle infected with bovine alphaherpesvirus types 1 and -5. To our knowledge, this is the first review analyzing the impact of perforin and granzymes on neurological infections caused by bovine herpesviruses.

Natural Killer Cell Group 7 Sequence in Cytotoxic Cells Optimizes Exocytosis of Lytic Granules Essential for the Perforin-Dependent, but Not Fas Ligand-Dependent, Cytolytic Pathway.

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Cytotoxic cells, such as CD8⁺ T cells or NK cells, have been shown to eliminate virus-infected cells or transformed cells primarily via two pathways: the perforin/granzyme-dependent pathway and the Fas ligand-Fas pathway; however, the precise cytolytic mechanisms have not been clarified thoroughly. In our previous study, we demonstrated that a T-box transcription factor, Eomesodermin (Eomes), may play important roles in activating the perforin pathway besides inducing perforin and granzyme B mRNA expression. In this study, we identified natural killer cell group 7 sequence (Nkg7), a molecule induced by Eomes, to be found critical for perforin-dependent cytotoxicity. Nkg7 mRNA expression in leukocytes from normal mice was mainly restricted to cells with cytotoxicity such as NK cells, NKT cells, and activated CD8⁺ T cells. The cytolytic activity of NK cells or CD8⁺ CTLs from Nkg7-deficient mice against Fas-negative target cells was reduced significantly, whereas Fas ligand-mediated cytotoxicity by Nkg7-deficient CTLs was not impaired. Further, translocation of granule membrane protein CD107a to the cell surface upon CD3 stimulation was defective in CD8⁺ CTLs from Nkg7 knockout, whereas surface induction of another granule membrane protein, CD63, was almost normal. In addition, analyses of lytic granules in CTLs by electron microscopy revealed that the number of lytic granules with dense cores was significantly reduced in Nkg7-knockout CTLs. These results indicate that Nkg7 may specifically contribute to efficient cytotoxicity via the perforin/granzyme pathway by enhancing the exocytosis of a particular type of lytic granules.

Escaping Death: How Cancer Cells and Infected Cells Resist Cell-Mediated Cytotoxicity.

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Cytotoxic lymphocytes are critical in our immune defence against cancer and infection. Cytotoxic T lymphocytes and Natural Killer cells can directly lyse malignant or infected cells in at least two ways: granule-mediated cytotoxicity, involving perforin and granzyme B, or death receptor-mediated cytotoxicity, involving the death receptor ligands, tumour necrosis factor-related apoptosis-inducing ligand (TRAIL) and Fas ligand (FasL). In either case, a multi-step pathway is triggered to facilitate lysis, relying on active pro-death processes and signalling within the target cell. Because of this reliance on an active response from the target cell, each mechanism of cell-mediated killing can be manipulated by malignant and infected cells to evade cytotoxic death. Here, we review the mechanisms of cell-mediated cytotoxicity and examine how cells may evade these cytotoxic processes. This includes resistance to perforin through degradation or reduced pore formation, resistance to granzyme B through inhibition or autophagy, and resistance to death receptors through inhibition of downstream signalling or changes in protein expression. We also consider the importance of tumour necrosis factor (TNF)-induced cytotoxicity and resistance mechanisms against this pathway. Altogether, it is clear that target cells are not passive bystanders to cell-mediated cytotoxicity and resistance mechanisms can significantly constrain immune cell-mediated killing. Understanding these processes of immune evasion may lead to novel ideas for medical intervention.

Granzymes and Mitochondria.

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Cytotoxic T lymphocytes and natural killer cells eliminate infected cells from the organism by triggering programmed cell death (apoptosis). The contents of the lytic granules of killer cells, including pore-forming proteins perforins and proteolytic enzymes granzymes, are released with the following

penetration of the released proteins into the target cells. Granzyme B initiates mitochondria-dependent apoptosis via (i) proapoptotic Bid protein, (ii) Mcl-1 and Bim proteins, or (iii) p53 protein. As a result, cytochrome c is released from the mitochondria into the cytoplasm, causing formation of apoptosomes that initiate the proteolytic cascade of caspase activation. Granzymes M, H, and F cause cell death accompanied by the cytochrome c release from the mitochondria. Granzyme A induces generation of reactive oxygen species (ROS), which promotes translocation of the endoplasmic reticulum-associated SET complex to the nucleus where it is cleaved by granzyme A, leading to the activation of nucleases that catalyze single-strand DNA breaks. Granzymes A and B penetrate into the mitochondria and cleave subunits of the respiratory chain complex I. One of the complex I subunits is also a target for caspase-3. Granzyme-dependent damage to complex I leads to the ROS generation and cell death.

Lymphocyte-mediated cytotoxicity.

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Virtually all of the measurable cell-mediated cytotoxicity delivered by cytotoxic T lymphocytes and natural killer cells comes from either the granule exocytosis pathway or the Fas pathway. The granule exocytosis pathway utilizes perforin to traffic the granzymes to appropriate locations in target cells, where they cleave critical substrates that initiate DNA fragmentation and apoptosis; granzymes A and B induce death via alternate, nonoverlapping pathways. The Fas/FasL system is responsible for activation-induced cell death but also plays an important role in lymphocyte-mediated killing under certain circumstances. The interplay between these two cytotoxic systems provides opportunities for therapeutic interventions to control autoimmune diseases and graft vs. host disease, but oversuppression of these pathways may also lead to increased viral susceptibility and/or decreased tumor cell killing.

The cytotoxic protease granzyme M is expressed by lymphocytes of both the innate and adaptive immune system.

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The cytotoxic serine protease granzyme M (GrM) is one of the five human granzymes, which are mainly expressed by cytotoxic T lymphocytes and/or NK cells. Upon perforin-dependent entry into a target cell, GrM cleaves specific substrates resulting in the onset of a unique cell death mechanism. However, the role of GrM in pathophysiological conditions is not clear yet. Knowledge of the expression and regulation of GrM by lymphocyte populations is instrumental for a better understanding of the contribution of this unique granzyme in health and disease. Two previous studies demonstrated GrM protein expression by lymphocytes of the innate immune system, i.e., NK cells, NKT cells, and gammadelta T cells, whereas its expression by CD8(+) T cells remained controversial. In the present study, we have investigated the expression and regulation of GrM in lymphocyte subsets in more detail. Flow cytometry analysis with a novel specific antibody against human GrM confirmed high expression of this protease by NK cells, NKT cells, and gammadelta T cells. CD8(+) T cells also expressed GrM and comparing the naive to early effector-memory, to late effector-memory, to effector subset, this expression gradually increased during differentiation. In contrast, CD4(+) T cells hardly expressed GrM. Quantitative PCR analysis for GrM mRNA levels in the diverse lymphocyte sub-populations confirmed the FACS results. GrM protein expression by lymphocyte populations was not significantly affected by a panel of GrB-inducing cytokines, indicating

that GrM expression is differentially regulated as compared to GrB. In conclusion, the human cytotoxic protease GrM is, besides by innate immune cells, also expressed by CD8(+) effector T cells, in particular by the differentiated effector CD27(-) CD45RO(-) subset. Our current findings support not only a role for GrM in the innate but also in the adaptive immune response.

DDVP markedly decreases the expression of granzyme B and granzyme 3/K in human NK cells.

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Natural killer (NK), lymphokine-activated killer (LAK), and cytotoxic T lymphocyte (CTL) cells kill target cells by the directed release of cytolytic granules that contain perforin, granzymes, and granulysin. We have found previously that dimethyl 2,2-dichlorovinyl phosphate (DDVP, dichlorvos), an organophosphorus pesticide, significantly decreased the expression of perforin, granzyme A (GrA), and granulysin and inhibited NK, LAK, and CTL activities. To further explore the mechanism of organophosphorus pesticide-induced inhibition of cell-mediated cytotoxicity, we examined whether organophosphorus pesticides affect the expression of GrB and Gr3/K in NK cells. We used an interleukin-2 (IL-2) independent human NK cell line, NK-92CI. We confirmed that NK-92CI cells express intracellular GrB and Gr3/K by flow cytometry, that NK-92CI cells are highly cytotoxic to K562 cells with a chromium release assay, and that DDVP significantly inhibited cytotoxic activity of NK-92CI cells in a dose- and time-dependent manner. We found that DDVP significantly decreased the expression of GrB and Gr3/K in NK-92CI cells in a dose- and time-dependent manner by flow cytometry. Immunocytochemical results showed that DDVP significantly decreases the level of GrB positive granules in NK-92CI cells, which may be due to degranulation. Taken together, DDVP significantly inhibits NK activity and reduces the intracellular GrB and Gr3/K levels in NK cells.
